

CE/CZ4055: CPS Security
Tutorial 5 - Microprocessor Security

- (1) A Fun Exercise: Just try to spot the differences between the following pictures. How do you compare this with micro-architectural cache attacks?



- (2) Assume that two user processes running on different cores of the same processor share the same memory stack. One is a malicious user while the other is the victim. Both the attacker and the victim each have access to their own segments in the main memory. Assume that the maximum capacity of the cache is 8 bytes. The attacker first fills the entire cache with some of his elements. The victim then performs an operation of table look up from the memory. The victim performs a look up of two bytes from his own main memory. After the victim's operation, what should the attacker do? What can he do to learn something about the victim's operation?
- (3) Assume that you can measure the power consumption of a device during its operation. Assume that the device leaks in the hamming weight model as follows: If the operated data is X , then the corresponding power consumption $P(X)$ can be modelled as

$$P(X) = HW(X) + \mathcal{D}_{0,1}$$

where $HW(X)$ denotes the Hamming weight of X and $\mathcal{D}_{0,1}$ denotes a Gaussian distribution with mean 0 and variance 1. The device is reading data from the memory and constantly storing in the registers. The memory has a data bus width of 4 bits and the power consumption trace observed has the values (3.9, 2.15, 1.8, 0.1, 2.2, 4.01, 1.6, 1.4). Now, based on the knowledge of the power leakage model and the power traces, what is your advantage in finding out the actual values that are read from the memory. Do you achieve better than just random guessing?